

GKE- Container Restart Count

PSD2Outbound GCO Alert - Container Restart count

Why do we have this alert?

- To ensure service is healthy

What does this alert mean?

- Pod is in pending state and is waiting to get scheduled
- Pod successful lifecycle: Pending -> Running -> Succeeded

Impact to customer

- With lesser number of pods, service performance may degrade. End user may experience slow response.

Investigation Steps

- In GCP console click on "Navigation Menu" "Kubernetes Cluster" "Clusters" "[outbound-prd-gke](#)"

Revision	Name	Status	Summary	Created on	Pods running/Pods total
7	x12a-account-service-0f8fbdbb-5wvft	OK	main: artifactory.sdlc.rft.gcp.db.com/for-public-local.com/ib/payment/service/directive/x12a-account-service-0.2.2-RELEASE	11 Jun 2024, 18:38:32	3/3

Revision	Name	Status	Restarts	Created on
7	x12a-account-service-0f8fbdbb-5wvft	Running	0	11 Jun 2024, 18:38:32
7	x12a-account-service-0f8fbdbb-5wvft	Running	0	14 Jun 2024, 06:44:02
7	x12a-account-service-0f8fbdbb-5wvft	Running	0	14 Jun 2024, 06:44:04

Name	Type	Endpoints
x12a-account-service	Cluster IP	240.20.190.170

- Navigate to the "Managed pods" section within the workload details. Identify pods with a high number in the "Restarts" column (anything more than 0 or 1 is a concern). Observe the "Status" column: Pending for too long, or CrashLoopBackOff are strong indicators of problems.

Revision	Name	Status	Restarts	Created on
43	crypto-service-6597c4bdc-b6gtr	Running	0	6 Dec 2023, 06:43:14
43	crypto-service-6597c4bdc-4p2fh	Running	0	6 Dec 2023, 06:47:49
43	crypto-service-6597c4bdc-b6gga	Running	0	6 Dec 2023, 07:51:36

- Select a problematic pod to investigate further and On the chosen pod's details page, go to the "Logs" tab for the primary container. Adjust the log viewer's time range to focus on the period just before the restarts.
- Scan logs for critical **keywords: ERROR, FATAL, EXCEPTION, OOM, Killed, failed, could not.**

Google Cloud db-prd-mp-psd2ce-obj

Search (/) for resources, docs, products and more

Kubernetes engine Pod details

crypto-service-6597c4b8c-b5sjv

Details Events Logs App errors (0) YAML

Container logs

Severity: Default Filter Search all fields and values

Scanned up to 29/01/2026, 15:26. Scanned 731.1 KB

2026-01-29 15:26:29.788 IST 2026-01-29 09:56:29 INFO [tp1978471802-22][auditId=12e351a8-67ed-4129-8293-b56bfead7bd, reqId=fef8bd4-67b-4c37-9d28-cca154cf9282, txId=b42cb9cb8756e827ab291ad3b407ab] SubscriptionService: 162 - Validating subscrip...

2026-01-29 15:26:29.788 IST 2026-01-29 09:56:29 INFO [tp1978471802-22][auditId=12e351a8-67ed-4129-8293-b56bfead7bd, reqId=fef8bd4-67b-4c37-9d28-cca154cf9282, txId=b42cb9cb8756e827ab291ad3b407ab] CSIService: 184 - v3 - Generating signature with...

2026-01-29 15:26:29.789 IST 2026-01-29 09:56:29 INFO [tp1978471802-22][auditId=12e351a8-67ed-4129-8293-b56bfead7bd, reqId=fef8bd4-67b-4c37-9d28-cca154cf9282, txId=b42cb9cb8756e827ab291ad3b407ab] CSIService: 323 - wrapping hash with algorithm ...

2026-01-29 15:26:29.795 IST 2026-01-29 09:56:29 INFO [tp1978471802-22][auditId=12e351a8-67ed-4129-8293-b56bfead7bd, reqId=fef8bd4-67b-4c37-9d28-cca154cf9282, txId=b42cb9cb8756e827ab291ad3b407ab] LoggingAspect: 61 - CSO PERFORMANCE LOG: executi...

2026-01-29 15:26:31.431 IST 2026-01-29 09:56:31 INFO [req-async-worker-8]() AuditLogService: 169 - PERF Audit log batch inserted size 3 time 11 batchSize : 6793376c-5215-4886-98d2-52f8e7bcb388

2026-01-29 15:26:31.758 IST POST 200 1.45 KB 15 ms Apache-HttpClient/5.5 (Java/17.0.16-1, http://crypto-service/v3/signhash

2026-01-29 15:26:31.764 IST 2026-01-29 09:56:31 INFO [tp1978471802-24][auditId=48e936c1-462f-46fe-af62-328579d5148e, reqId=f6f87db1-73a1-4dff-be19-208b56ff38fc, txId=8523483827f683b365528316918f6f3] SubscriptionService: 162 - Validating subscrip...

2026-01-29 15:26:31.764 IST 2026-01-29 09:56:31 INFO [tp1978471802-24][auditId=48e936c1-462f-46fe-af62-328579d5148e, reqId=f6f87db1-73a1-4dff-be19-208b56ff38fc, txId=8523483827f683b365528316918f6f3] CSIService: 184 - v3 - Generating signature with...

- If Container also restart click on the filter pods and Navigate to the "Container" and check the status if it is running then fine , else check the why it got restarted .

Google Cloud db-prd-mp-psd2ce-obj

Search (/) for resources, docs, products and more

Kubernetes engine Pod details

crypto-service-6597c4b8c-b5sjv

Details Events Logs App errors (0) YAML

Containers

Name	Status	Image	Restart count	Logs
istio-proxy	Running	gke.gcr.io/asmn/proxyv2.1.24.5-asm.3	0	View logs
main	Running	artifactory.sdlc.cil.gcp.db.com/dkr-public-local/com/db/payment/service/directive/crypto-service:0.93-RELEASE	0	View logs

Containers

Init containers

Volumes

workload-socket

credential-socket

workload-certs

istio-envoy

istio-data

istio-podinfo

istio-token

istiod-ca-cert

volume-crypto-configmap

volume-secrets-certificates

volume-crypto-secrets

volume-oracle-secrets

volume-crypto-secrets-keystore-and-truststore

kube-api-access-w5mrz

- To investigate further Navigate to "main" and click on view logs . and Scan logs for critical **keywords: ERROR, FATAL, EXCEPTION, OOM, Killed, failed, could not.**

Google Cloud db-prd-mp-psd2ce-obj

Search (/) for resources, docs, products and more

Logs Explorer

Query language

resource.type="k8s_container"

resource.labels.project_id="db-prd-2vm-mp-psd2ce-obj"

resource.labels.location="europe-west3"

resource.labels.cluster_name="outbound-prd-gke"

resource.labels.namespace_name="psd2ce-microservice"

resource.labels.pod_name="crypto-service-6597c4b8c-b5sjv"

resource.labels.container_name="main"

Sample queries

Query language guide

Fields

Search fields and values

System metadata

Severity

Cluster name

Container name

Timeline

103,932 results

2026-01-29 11:12:57 INFO [req-async-worker-5]() AuditLogService: 169 - PERF Audit log batch inserted size 2 time 23 batchSize : f0dbd71-4679-415e-956d-985158f95b9d

2026-01-29 16:42:59.188 [main] [POST] 200 1.45 KB 15 ms Apache-HttpClient/5.5 (Java/17.0.16-1, http://crypto-service/v3/signhash

2026-01-29 16:42:59.194 [main] [POST] 200 1.45 KB 15 ms Apache-HttpClient/5.5 (Java/17.0.16-1, http://crypto-service/v3/signhash

2026-01-29 11:12:59 INFO [tp1978471802-22][auditId=73b48fcb-bcf9-493a-966b-fd2cb6133ce, reqId=a24ace44-af69-4669-bab5-3ae998d4c9e7, txId=b57787d3df72a28f8c37bdc349bd7cd] SubscriptionService: 162 - Validating subscription.

2026-01-29 16:42:59.195 [main] [POST] 200 1.45 KB 15 ms Apache-HttpClient/5.5 (Java/17.0.16-1, http://crypto-service/v3/signhash

2026-01-29 16:42:59.195 [main] [POST] 200 1.45 KB 15 ms Apache-HttpClient/5.5 (Java/17.0.16-1, http://crypto-service/v3/signhash

2026-01-29 16:42:59.195 [main] [POST] 200 1.45 KB 15 ms Apache-HttpClient/5.5 (Java/17.0.16-1, http://crypto-service/v3/signhash

2026-01-29 16:42:59.283 [main] [POST] 200 1.45 KB 15 ms Apache-HttpClient/5.5 (Java/17.0.16-1, http://crypto-service/v3/signhash

2026-01-29 11:12:59 INFO [tp1978471802-22][auditId=73b48fcb-bcf9-493a-966b-fd2cb6133ce, reqId=a24ace44-af69-4669-bab5-3ae998d4c9e7, txId=b57787d3df72a28f8c37bdc349bd7cd] CSIService: 184 - v3 - Generating signature with hashAlgorithm : SHA-256 and signatureAlgorithm : NONEwithRSA

2026-01-29 11:12:59 INFO [tp1978471802-22][auditId=73b48fcb-bcf9-493a-966b-fd2cb6133ce, reqId=a24ace44-af69-4669-bab5-3ae998d4c9e7, txId=b57787d3df72a28f8c37bdc349bd7cd] CSIService: 323 - wrapping hash with algorithm : SHA-256

2026-01-29 11:12:59 INFO [tp1978471802-22][auditId=73b48fcb-bcf9-493a-966b-fd2cb6133ce, reqId=a24ace44-af69-4669-bab5-3ae998d4c9e7, txId=b57787d3df72a28f8c37bdc349bd7cd] LoggingAspect: 61

- Based on log analysis check with dev team for further action.

Verification Steps:

- after container restarts it looks running fine then no action required but it shows initialize state or any crashloop then do the proper log analysis and reach out to Dev team.